

Homework Submission

Student Name
September 22, 2026

Problem Statement

Solve the quadratic equation

$$x^2 + 4x + 4 = 0.$$

Key Principles and Formulas

- **Factoring:** A quadratic $ax^2 + bx + c$ can often be written as $(dx + e)^2$ when it is a perfect square.
- **Completing the Square:** Transform $ax^2 + bx + c$ into $(x + \frac{b}{2a})^2 = \text{constant}$ to isolate x .
- **Quadratic Formula:** For $ax^2 + bx + c = 0$,

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}.$$

Detailed Step-by-Step Solution

Method 1: Factoring / Completing the Square

$$\begin{aligned}x^2 + 4x + 4 &= 0 \\x^2 + 4x &= -4 \\x^2 + 4x + 4 &= -4 + 4 \\(x + 2)^2 &= 0 \\x + 2 &= 0 \\x &= -2\end{aligned}$$

Method 2: Quadratic Formula

$$\begin{aligned}a &= 1, & b &= 4, & c &= 4 \\ \Delta &= b^2 - 4ac = 4^2 - 4 \cdot 1 \cdot 4 = 16 - 16 = 0 \\ x &= \frac{-b \pm \sqrt{\Delta}}{2a} = \frac{-4 \pm \sqrt{0}}{2 \cdot 1} = \frac{-4}{2} \\ x &= -2\end{aligned}$$

Final Answer

$$\boxed{x = -2}$$

Study Notes and Key Takeaways

- A discriminant $\Delta = b^2 - 4ac$ equal to zero indicates a repeated (double) root.
- Recognize perfect-square trinomials: $x^2 + 4x + 4 = (x + 2)^2$; this speeds up solving.
- When completing the square, always add the same term to both sides to maintain equality.